NIRVANA – Master Specification
Volume 10: Frontend Architecture & Application Structure

Objective

Define a scalable frontend architecture that delivers a premium, responsive wealth-management experience while remaining modular and maintainable.

Technology Stack

Next.js (App Router), React, TypeScript, Tailwind CSS, Framer Motion, Recharts, React Hook Form, Zod validation and TanStack Query for server state.

Application Structure

Organize by feature modules: Authentication, Dashboard, Clients, Portfolio, Goals, Recommendations, Withdrawal Planner, AI Advisor, Reports, Admin and Shared UI.

Layout

Persistent sidebar, top navigation, breadcrumb trail, global search, notifications, profile menu and contextual action panels.

State Management

Use local component state where appropriate, global state for authenticated user/session and TanStack Query for cached API data.

Routing

Protected routes for authenticated users with role-based access. Separate layouts for client, advisor and administrator dashboards.

Component Library

Reusable cards, KPI tiles, charts, forms, upload components, tables, filters, modals, dialogs, timelines, accordions and AI chat interface.

Forms

Multi-step onboarding with autosave, inline validation, progress indicators, draft recovery and dynamic field rendering based on user responses.

Performance

Lazy-load charts, virtualize large tables, optimize images, code split by route, cache API responses and provide skeleton loaders.

Accessibility

Keyboard navigation, WCAG-compliant contrast, semantic HTML, screen-reader labels, focus management and responsive typography.

Frontend–Backend Contract

All financial calculations originate from backend services. Frontend displays normalized data, renders visualizations and presents AI explanations without performing core financial logic.

Future Expansion

Architecture should support mobile applications, multilingual interfaces, white-label deployments and advisor-specific workspaces with minimal refactoring.